KAIZEN MATHS

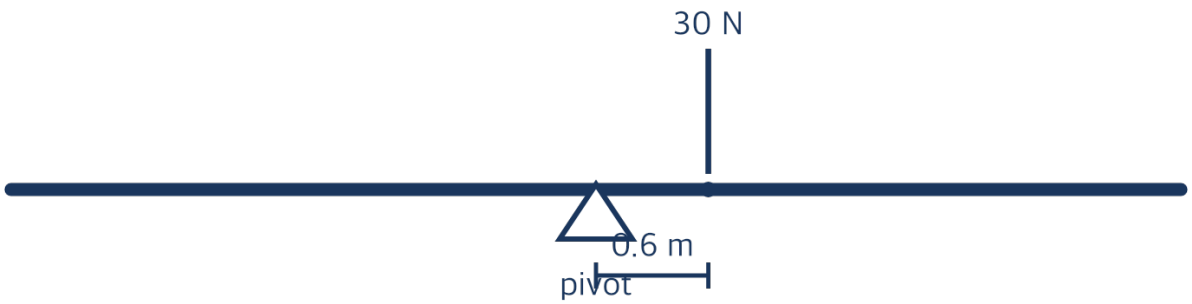
Lesson starter

Moments and Equilibrium

Moment Basics

Name _____

SA1



Find the moment of the force about the pivot and state its direction.

MEC-4D51-L1 · SA

Show your working

KAIZEN MATHS

Lesson starter

Moments and Equilibrium

Moment Basics

Name _____

SB1



State whether the force produces a clockwise or anticlockwise moment about the pivot. Then calculate the size of the moment.

MEC-4D51-L1 · SB

Show your working

KAIZEN MATHS

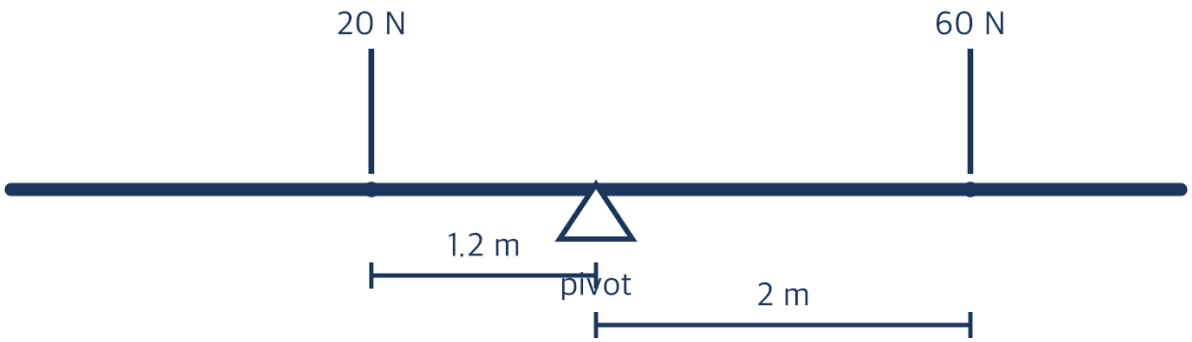
Lesson starter

Moments and Equilibrium

Moment Basics

Name _____

SC1



Find the resultant moment about the pivot.

MEC-4D51-L1 · SC

Show your working

KAIZEN MATHS

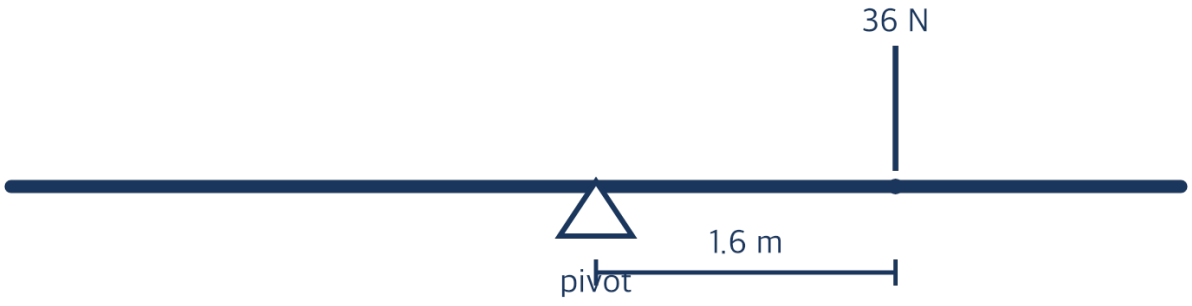
Lesson starter

Moments and Equilibrium

Moment Basics

Name _____

SD1



Find the moment of the force about the pivot and state its direction.

MEC-4D51-L1 · SD

Show your working

KAIZEN MATHS

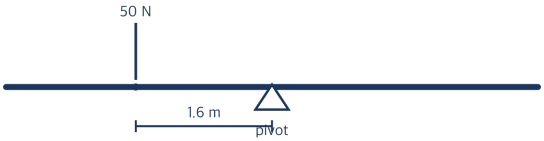
Exit ticket

Moments and Equilibrium

Moment Basics

Name _____

EA1



State whether the force produces a clockwise or anticlockwise moment about the pivot. Then calculate the size of the moment.

Show your working

KAIZEN MATHS

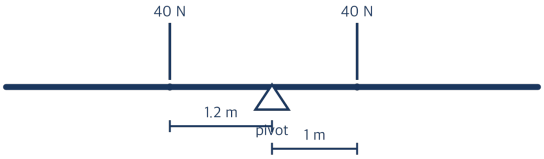
Exit ticket

Moments and Equilibrium

Moment Basics

Name _____

EB1



Find the resultant moment about the pivot.

Show your working

KAIZEN MATHS

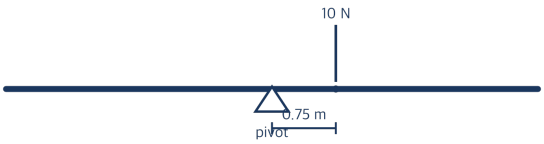
Exit ticket

Moments and Equilibrium

Moment Basics

Name _____

EC1



Find the moment of the force about the pivot and state its direction.

Show your working

KAIZEN MATHS

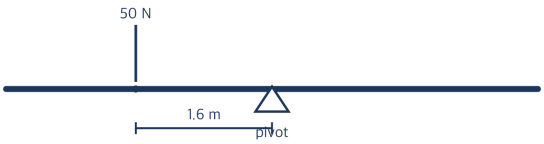
Exit ticket

Moments and Equilibrium

Moment Basics

Name _____

EA1



State whether the force produces a clockwise or anticlockwise moment about the pivot. Then calculate the size of the moment.

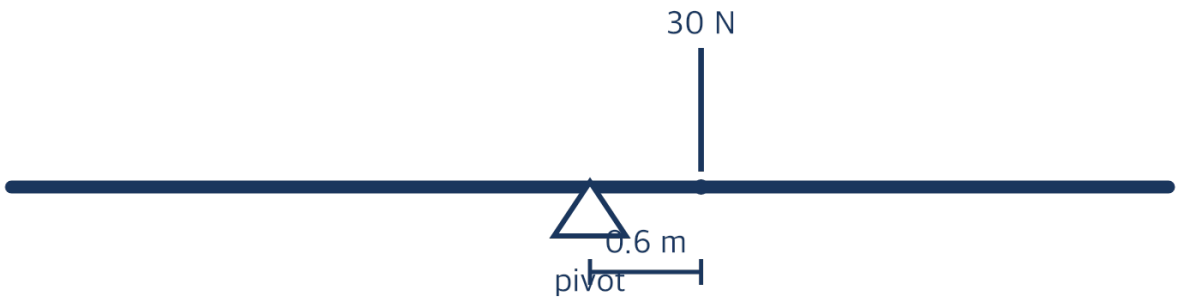
Show your working

Teacher answers & working

Moments and Equilibrium · Moment Basics

MEC-4D51-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Find the moment of the force about the pivot and state its direction.

Answer:

18 N m clockwise

Working:

Identify the force and perpendicular distance from the pivot.

force = 30 N, distance = 0.6 m

Use moment = force × perpendicular distance.

moment = 30 × 0.6 = 18 N m

Decide the turning direction from the diagram.

A downward force on the right of the pivot turns the beam clockwise.

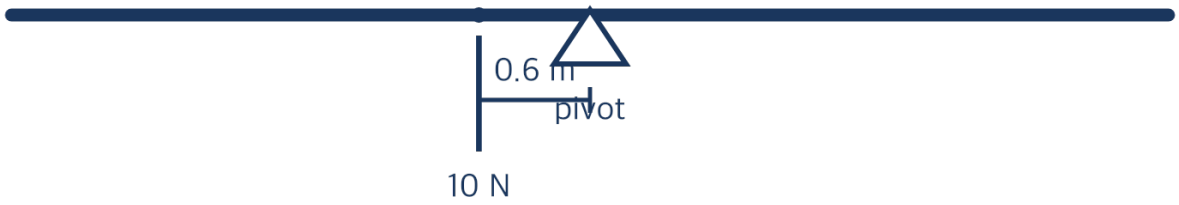
The moment is 18 N m clockwise.

Teacher answers & working

Moments and Equilibrium · Moment Basics

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SB1



State whether the force produces a clockwise or anticlockwise moment about the pivot. Then calculate the size of the moment.

Answer:

clockwise, 6 N m

Working:

Look at which side of the pivot the force acts on and which way the force points.

The force acts on the left of the pivot and points upwards.

Decide the turning direction.

This produces an clockwise moment about the pivot.

Calculate the size of the moment.

moment = 10 × 0.6 = 6 N m

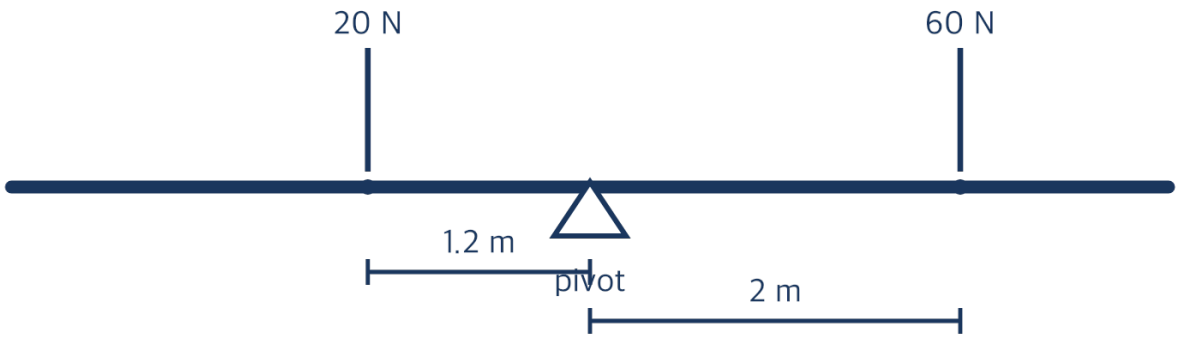
The moment is clockwise with size 6 N m.

Teacher answers & working

Moments and Equilibrium · Moment Basics

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SC1



Find the resultant moment about the pivot.

Answer:

96 N m clockwise

Working:

Calculate the anticlockwise moment from the left-hand force.

anticlockwise moment = 20 × 1.2 = 24 N m

Calculate the clockwise moment from the right-hand force.

clockwise moment = 60 × 2 = 120 N m

Subtract the smaller moment from the larger moment.

resultant moment = |120 - 24| = 96 N m

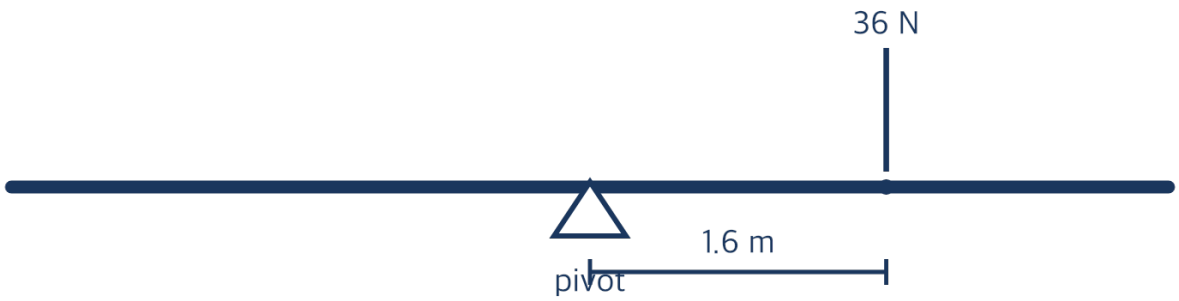
The resultant moment is 96 N m clockwise.

Teacher answers & working

Moments and Equilibrium · Moment Basics

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SD1



Find the moment of the force about the pivot and state its direction.

Answer:

57.6 N m clockwise

Working:

Identify the force and perpendicular distance from the pivot.

force = 36 N, distance = 1.6 m

Use moment = force × perpendicular distance.

moment = 36 × 1.6 = 57.6 N m

Decide the turning direction from the diagram.

A downward force on the right of the pivot turns the beam clockwise.

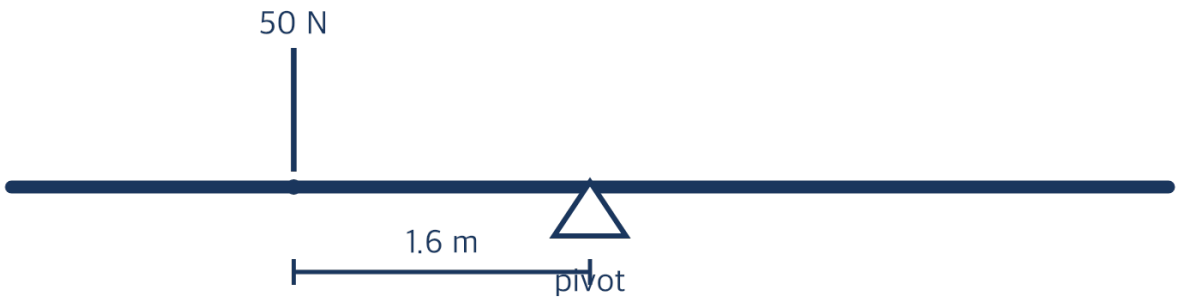
The moment is 57.6 N m clockwise.

Teacher answers & working

Moments and Equilibrium · Moment Basics

MEC-4D51-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



State whether the force produces a clockwise or anticlockwise moment about the pivot. Then calculate the size of the moment.

Answer:

anticlockwise, 80 N m

Working:

Look at which side of the pivot the force acts on and which way the force points.

The force acts on the left of the pivot and points downwards.

Decide the turning direction.

This produces an anticlockwise moment about the pivot.

Calculate the size of the moment.

moment = 50 × 1.6 = 80 N m

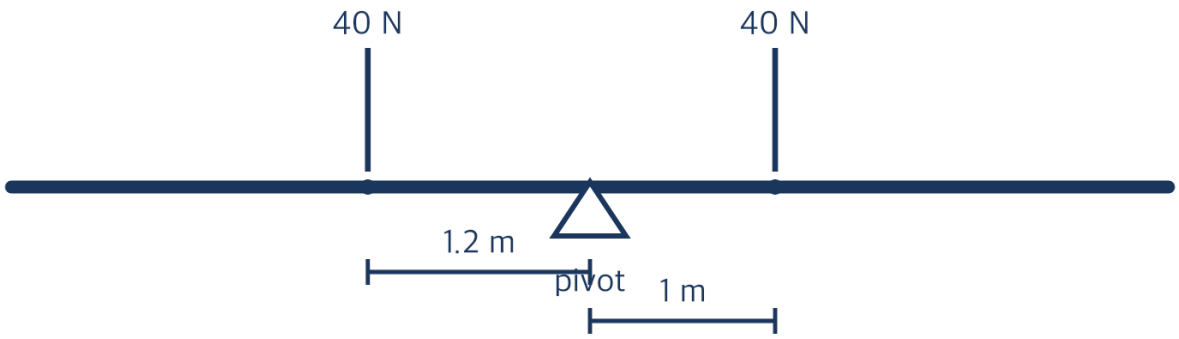
The moment is anticlockwise with size 80 N m.

Teacher answers & working

Moments and Equilibrium · Moment Basics

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EB1



Find the resultant moment about the pivot.

Answer:

8 N m anticlockwise

Working:

Calculate the anticlockwise moment from the left-hand force.

anticlockwise moment = 40 × 1.2 = 48 N m

Calculate the clockwise moment from the right-hand force.

clockwise moment = 40 × 1 = 40 N m

Subtract the smaller moment from the larger moment.

resultant moment = |40 - 48| = 8 N m

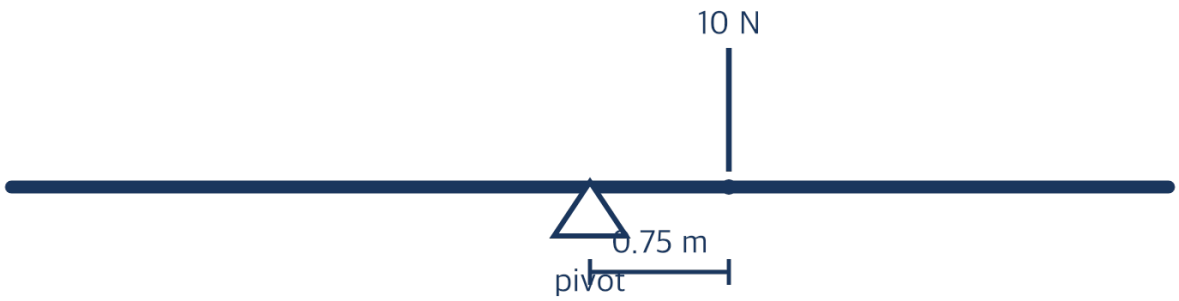
The resultant moment is 8 N m anticlockwise.

Teacher answers & working

Moments and Equilibrium · Moment Basics

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EC1



Find the moment of the force about the pivot and state its direction.

Answer:

7.5 N m clockwise

Working:

Identify the force and perpendicular distance from the pivot.

force = 10 N, distance = 0.75 m

Use moment = force × perpendicular distance.

moment = 10 × 0.75 = 7.5 N m

Decide the turning direction from the diagram.

A downward force on the right of the pivot turns the beam clockwise.

The moment is 7.5 N m clockwise.